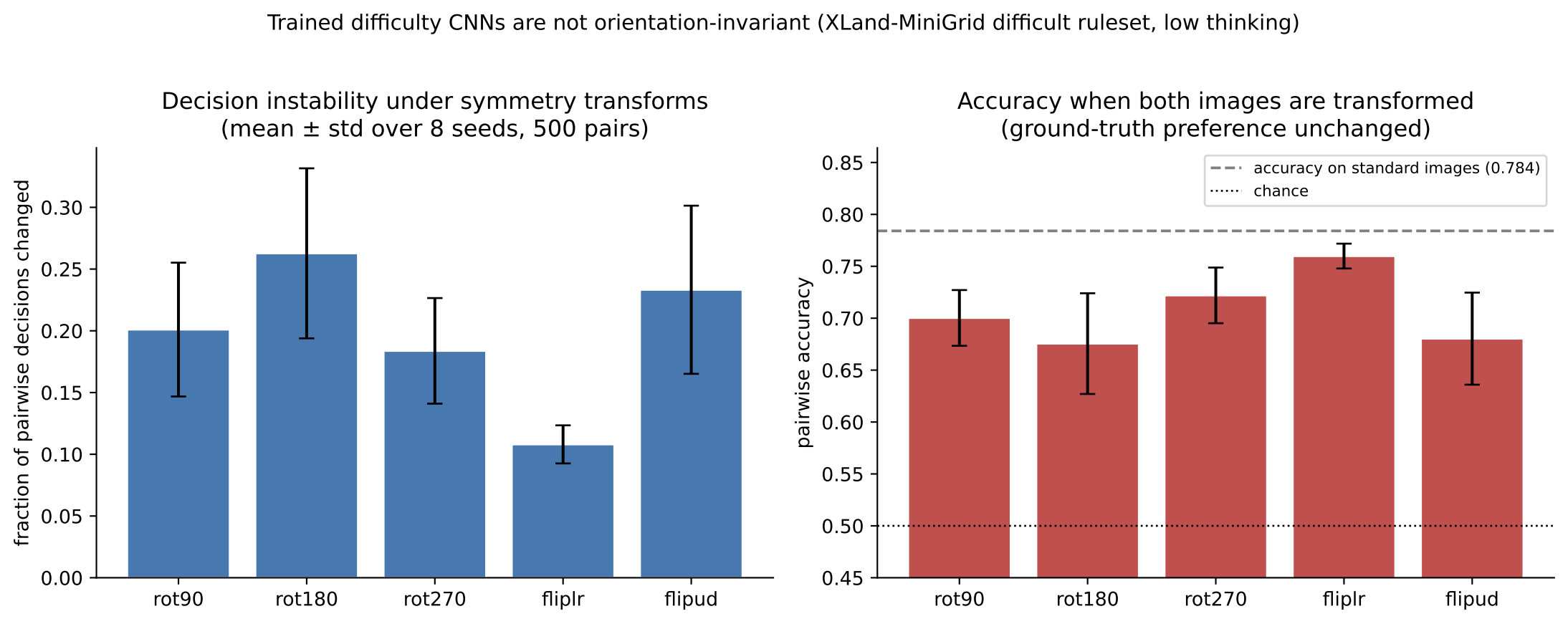
Trained difficulty CNNs are not orientation-invariant (XLand-MiniGrid difficult ruleset, low thinking)
Decision instability under symmetry transforms
(mean ± std over 8 seeds, 500 pairs)
fraction of pairwise decisions changed
0.00
0.05
0.10
0.15
0.20
0.25
0.30
rot90
rot180
rot270
fliplr
flipud
Accuracy when both images are transformed
(ground-truth preference unchanged)
pairwise accuracy
0.45
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
accuracy on standard images (0.784)
chance
rot90
rot180
rot270
fliplr
flipud